

Comparative Analysis Report

Model Selection and Hyperparameter Tuning

1. Introduction

This project investigates model selection and hyperparameter optimization for a binary classification problem. Random Forest, Gradient Boosting, and Support Vector Machine (SVM) were evaluated using GridSearchCV and RandomizedSearchCV.

Models were compared using accuracy, precision, recall, F1-score, cross-validation F1-score, validation F1-score, and training time. F1-score was used as the primary optimization metric because the target classes are imbalanced.

2. Dataset and Problem Definition

The wine-quality dataset was converted into a binary classification problem: Quality ≥ 7 was labelled Good and Quality < 7 was labelled Not Good.

Class	Samples
Not Good	208
Good	32
Total	240

The test set is substantially imbalanced toward the Not Good class. Consequently, accuracy alone does not fully describe model performance.

3. Models Evaluated

3.1 Random Forest

Random Forest combines multiple decision trees to produce predictions. Hyperparameters explored included `n_estimators`, `max_depth`, `min_samples_split`, and `min_samples_leaf`.

3.2 Gradient Boosting

Gradient Boosting builds trees sequentially, with each new tree attempting to correct previous errors. Hyperparameters included `n_estimators`, `learning_rate`, `max_depth`, and `min_samples_split`.

3.3 Support Vector Machine

SVM identifies an effective decision boundary between classes. `C`, `kernel`, and `gamma` were explored, with feature standardization included in the SVM pipeline.

4. Hyperparameter Optimization

4.1 GridSearchCV

GridSearchCV exhaustively evaluates the specified parameter combinations using 5-fold cross-validation and F1-score.

Model	Candidates	CV Folds	Total Fits
Random Forest	24	5	120
Gradient Boosting	24	5	120
SVM	12	5	60

4.2 RandomizedSearchCV

RandomizedSearchCV samples 20 configurations per model using 5-fold cross-validation. Each model therefore performs 100 fits. The scoring metric was F1-score and `random_state` was 42.

5. GridSearchCV Performance

Model	Best CV F1	Validation F1	Time
Random Forest	0.5001	0.6923	58.22 s
Gradient Boosting	0.5208	0.7037	59.19 s
SVM	0.4511	0.6538	2.78 s

Gradient Boosting achieved the highest GridSearchCV CV F1-score (0.5208) and validation F1-score (0.7037). SVM was fastest but had the lowest validation F1-score.

6. RandomizedSearchCV Performance

Model	Best CV F1	Validation F1	Time
Random Forest	0.4593	0.7037	46.16 s
Gradient Boosting	0.5282	0.6909	58.56 s
SVM	0.4892	0.5818	71.85 s

Gradient Boosting achieved the highest randomized-search CV F1-score (0.5282), while Random Forest achieved the highest validation F1-score (0.7037) and the shortest randomized-search time.

7. GridSearchCV vs RandomizedSearchCV

Model	Grid CV	Random CV	Grid Val.	Random Val.
Random Forest	0.5001	0.4593	0.6923	0.7037
Gradient Boosting	0.5208	0.5282	0.7037	0.6909
SVM	0.4511	0.4892	0.6538	0.5818

The better search strategy depended on the model. RandomizedSearchCV improved Random Forest validation performance, GridSearchCV produced better Gradient Boosting validation performance, and GridSearchCV clearly outperformed RandomizedSearchCV for SVM validation performance.

8. Final Test-Set Performance

Model	Accuracy	Precision	Recall	F1-score
Random Forest	0.9292	0.8947	0.5312	0.6667
Gradient Boosting	0.9125	0.7037	0.5938	0.6441
SVM	0.8833	0.6000	0.3750	0.4615

9. Detailed Performance

Random Forest

Accuracy: 92.92%. Precision: 89.47%. Recall: 53.12%. F1-score: 66.67%. Random Forest achieved the highest accuracy, precision, and F1-score.

Gradient Boosting

Accuracy: 91.25%. Precision: 70.37%. Recall: 59.38%. F1-score: 64.41%. Gradient Boosting achieved the highest recall for the Good class.

SVM

Accuracy: 88.33%. Precision: 60.00%. Recall: 37.50%. F1-score: 46.15%. SVM produced the lowest overall test performance.

10. Class-Level Performance

Model / Class	Precision	Recall	F1	Support
RF - Not Good	0.93	0.99	0.96	208
RF - Good	0.89	0.53	0.67	32
GB - Not Good	0.94	0.96	0.95	208
GB - Good	0.70	0.59	0.64	32
SVM - Not Good	0.91	0.96	0.93	208
SVM - Good	0.60	0.38	0.46	32

11. Model Performance Across Configurations

Model	Configuration	CV F1	Validation F1
Random Forest	GridSearchCV	0.5001	0.6923
Random Forest	RandomizedSearchCV	0.4593	0.7037
Gradient Boosting	GridSearchCV	0.5208	0.7037
Gradient Boosting	RandomizedSearchCV	0.5282	0.6909
SVM	GridSearchCV	0.4511	0.6538
SVM	RandomizedSearchCV	0.4892	0.5818

12. Training Time

Search	Random Forest	Gradient Boosting	SVM
GridSearchCV	58.22 s	59.19 s	2.78 s
RandomizedSearchCV	46.16 s	58.56 s	71.85 s

13. Effect of Class Imbalance

The test set contains 208 Not Good observations and 32 Good observations, meaning approximately 86.7% of the test samples belong to the Not Good class. This imbalance can produce high accuracy even when minority-class recall is lower. For this reason, F1-score, precision, and recall were considered alongside accuracy.

14. Final Model Selection

Random Forest was selected as the final model because it achieved the highest test accuracy (92.92%), highest precision (89.47%), and highest F1-score (66.67%). Gradient Boosting had higher recall (59.38%), but Random Forest provided the strongest overall balance according to the primary F1-based evaluation objective.

15. Overall Ranking

Rank	Model	Accuracy	Precision	Recall	F1-score
1	Random Forest	92.92%	89.47%	53.12%	66.67%
2	Gradient Boosting	91.25%	70.37%	59.38%	64.41%
3	SVM	88.33%	60.00%	37.50%	46.15%

16. Key Findings

- Random Forest was the best overall model based on test F1-score and accuracy.
- Gradient Boosting achieved the highest recall for the Good class.
- SVM produced the lowest overall predictive performance.

- GridSearchCV and RandomizedSearchCV produced different optimal configurations.
- RandomizedSearchCV was more efficient for Random Forest in this experiment.
- Accuracy alone was insufficient because of class imbalance.
- F1-score provided a more balanced measure of performance.
- Tree-based ensemble models outperformed SVM on the final test set.

17. Limitations

- Class imbalance makes minority-class recall more difficult.
- The dataset contains a limited number of minority-class observations.
- The tested hyperparameter ranges represent only a subset of all possible configurations.
- Search iterations were limited to maintain reasonable execution time.
- Results may vary with a different train/validation/test split or independent dataset.

18. Conclusion

The experiment demonstrated systematic hyperparameter optimization using GridSearchCV and RandomizedSearchCV across Random Forest, Gradient Boosting, and SVM. Random Forest achieved the best overall test performance with 92.92% accuracy, 89.47% precision, 53.12% recall, and 66.67% F1-score.

Gradient Boosting was a strong alternative and achieved the highest recall for the Good class. SVM produced the lowest overall predictive performance. Considering class imbalance and the need to balance precision and recall, Random Forest was selected as the final model.